

26 An alternating potential difference is connected across a pure resistor and the frequency f of the supply is varied, keeping the r.m.s voltage constant. The mean rate of production of heat in the resistor is

- A** proportional to f
- B** proportional to $f^{1/2}$
- C** inversely proportional to f
- D** independent of f

Ans: D

Mean rate of production of heat, $P = \frac{V_{rms}^2}{R}$

It is independent of f .